

Scalability & Future Plan

Date	15/07/2024
Team ID	SWTID-2026-3095
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Only 1.69% of training data is high-risk, capping ROC-AUC around 0.71-0.75 regardless of tuning	Limits how confidently the model alone can rank risk	Medium
2	No persistent audit-trail database — decisions are not stored or reviewable after the fact	Analysts can't look back at past decisions or track outcomes over time	High
3	Single Flask dev server, not a production WSGI/ASGI deployment	Not resilient to real concurrent traffic or process crashes	High
4	No authentication or role-based access control	Anyone with network access can submit or view applications	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-process Flask dev server, in-process test client only	Deploy behind Gunicorn/Nginx (or Render) with multiple worker processes
2	Data Storage	Flat.pkl/.json files, no database	Add PostgreSQL for an audit trail of applications and decisions
3	Performance	~15ms average per full ML+SHAP prediction, single process	Add response caching for repeated SHAP background computations; async request handling
4	Security	No authentication; no encryption of applicant data at rest	Add analyst login (JWT/OAuth2) and encrypt any persisted applicant data

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Persistent audit-trail database (PostgreSQL) + analyst authentication	Aug 2026	Enables historical review and accountability for every decision
Phase 3	Calibrated probability outputs; real credit-bureau API integration; model drift monitoring	Q4 2026	More trustworthy scores and early warning if real-world data shifts from training data
Phase 4	Containerized deployment (Docker) on Render/AWS with CI/CD	Q4 2026	Production-grade hosting, reproducible deployments, automated testing on every commit